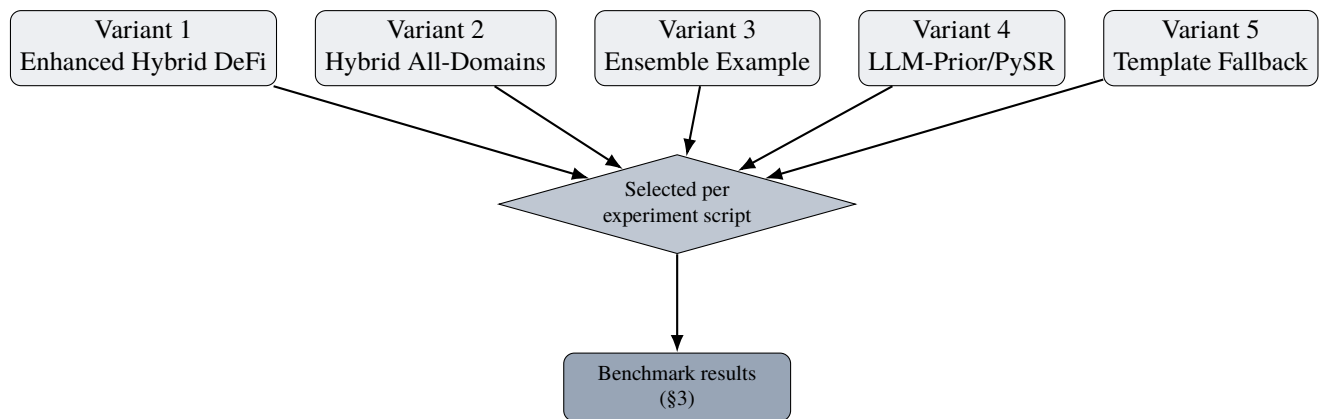


HypatiaX Hybrid Variants

Naming, Experiment Cross-Reference & Algorithm Diagrams

Source repository: <https://github.com/sednabcn/LLM-HypatiaX-REPRO> (master)

Built on the prior static-analysis report (core/, experiments/, tools/, protocols/)



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1 1) Variant list and naming

Five hybridisation *variants* exist across `hypatiax/core/` and `hypatiax/tools/`. Each implements a different way of combining an LLM-generated symbolic prior with a numeric predictor (neural network, genetic-programming symbolic regression, or a template library). Table 1 names each one and gives its defining class, its combination strategy, and where it lives in the repository.

#	Name	Defining class / function	Combination strategy
1	Enhanced Hybrid DeFi	<code>EnhancedHybridSystemDeFi</code>	Fixed $R^2 \geq 0.85 + \text{margin} > 0.05$ threshold gate, after a <code>scipy.optimize.curve_fit</code> constant refit of the LLM formula
2	Hybrid All-Domains	<code>HybridSystemAllDomains</code>	Tie-favours-symbolic gate ($r_{llm}^2 \geq r_{nn}^2$), with a <code>force_llm</code> override that skips the NN for known power-law domains
3	Ensemble Example	<code>ensemble_llm_nn()</code>	Precision-weighted blend: $\text{weight} \propto R^2 / \sigma_{resid}$ (inverse-residual-std), no hard gate
4	LLM-Prior / PySR Family	<code>SymbolicEngineWithLLM</code> (4 modes: <code>none/seed/hybrid/fallback</code>), driven by <code>HybridDiscoverySystem</code>	LLM and genetic-programming (PySR) take turns proposing/refining one candidate expression
5	Template Physics Fallback	<code>PhysicsAwareRegressor</code> (template-library) / <code>SmartStructureDetector</code> (template-free)	Fits a library of domain-specific forms (Michaelis-Menten, Bernoulli, ...) or infers additive/multiplicative structure statistically, used when Variant 4 underperforms

Table 1: The five HypatiaX hybrid variants.

1.1 Where each variant lives

[C1] `hypatiax/core/generation/hybrid_defi_system/hybrid_system_nn_defi_domain.py` – Variant 1

[C2] `hypatiax/core/generation/hybrid_all_domains_llm_nn/hybrid_system_llm_nn_all_domains.py` – Variant 2

[C3] `hypatiax/core/generation/hybrid_defi_llm_nn/hybrid_ensemble_system_defi_domain.py` – Variant 3

[T1]/[T2] `hypatiax/tools/symbolic/hybrid_system_v50_2.py` (orchestrator) / `symbolic_engine.py` (mode logic) – Variant 4

[T3]/[T4] `hypatiax/tools/symbolic/physics_aware_regressor.py` / `smart_structure_detector.py` – Variant 5

2 2) Variant → experiment cross-reference

Table 2 links each variant to the experiment scripts in `hypatiaux/experiments/benchmarks/` and `hypatiaux/experiments/` that actually construct or import it, confirmed by direct source inspection (class names / imports found in each file).

Variant	Experiment file(s)	Evidence (import / usage found)
1. Enhanced Hybrid DeFi	<code>hypatiaux_defi_benchmark_v4.py</code> (run directly by the <code>exp1/exp1b</code> pipeline steps, Tab 9-15), <code>run_comparative_suite_benchmark_v2.py</code> , <code>_pca.py</code> , <code>run_noise_sweep_benchmark.py</code> , <code>run_sample_complexity_benchmark.py</code> , <code>test_enhanced_defi_extrapolation.py</code>	<code>run_all.sh</code> 's <code>exp1/exp1b</code> steps invoke <code>hypatiaux_defi_benchmark_v4.py</code> directly as the paper's core DeFi benchmark (which itself calls <code>EnhancedHybridSystemDeFi</code>); <code>run_noise_sweep_benchmark.py</code> and <code>run_sample_complexity_benchmark.py</code> name it explicitly as "method 3"
2. Hybrid All-Domains	<code>hybrid_system_llm_nn_all_domains.py</code> (run directly as step (§10.9) <code>cds</code> into <code>hybrid_all_domains</code> pipeline step), <code>run_sample_complexity_benchmark.py</code> , <code>run_comparative_suite_benchmark_v2.py</code>	<code>run_all.sh</code> 's <code>hybrid_all_domains</code> step (§10.9) <code>cds</code> into <code>hybrid_all_domains_llm_nn/</code> and runs <code>hybrid_system_llm_nn_all_domains.py</code> directly – the clearest evidence of any variant, since this is Variant 2's own defining file executed as the experiment; <code>run_sample_complexity_benchmark.py</code> names <code>HybridSystemLLMNN</code> <code>all-domains</code> alongside Variant 1
3. Ensemble Example	<code>hypatiaux_defi_benchmark_v3.py</code> , <code>_v4.py</code> , <code>_v3_pca.py</code> , <code>_v4_pca.py</code> , <code>test_enhanced_defi_extrapolation.py</code>	Each defines/calls a local <code>_ensemble_llm_nn()</code> function; <code>test_enhanced_defi_extrapolation.py</code> imports <code>ensemble_llm_nn</code> directly and calls it – so, despite being introduced as a "worked example", this variant is exercised in the DeFi v3/v4 benchmark family and its tests
4. LLM-Prior / PySR Family	<code>hypatia.py</code> , <code>exp1_ablation.py</code> (run by pipeline step <code>exp1_ablation</code> , Tab 5, §10.6), <code>exp1_five_system.py</code> (step <code>exp1_five</code> , Tab 1, §10.1), <code>exp3_nguyen12_consolidated.py</code> (step <code>exp3</code> , §10.8), <code>run_comparative_suite_benchmark_v2.py</code>	<code>hypatia.py</code> defines <code>get_llm_prior()</code> directly; <code>exp1_ablation.py</code> and <code>exp1_five_system.py</code> import <code>HybridDiscoverySystem</code> (which wraps <code>SymbolicEngineWithLLM</code>) and are run by <code>run_all.sh</code> with <code>ANTHROPIC_API_KEY</code> required; <code>exp3_nguyen12_consolidated.py</code> imports <code>get_llm_prior</code> from <code>hypatia</code> ; the comparative-suite scripts import <code>SymbolicEngineWithLLM</code> and <code>HybridDiscoverySystem</code> by name

Variant	Experiment file(s)	Evidence (import / usage found)
5. Template Physics Fallback	exp1_ablation.py, run_comparative_suite_benchmark_v2.py/crpt.py (indirectly, via HybridDiscoverySystem)	Not imported directly by any experiment script, reached only as an internal fallback tier inside HybridDiscoverySystem [T1] when its PySR/LLM search underperforms – so it is exercised only transitively, wherever Variant 4 is exercised

Table 2: Variant → experiment mapping, from direct source inspection.

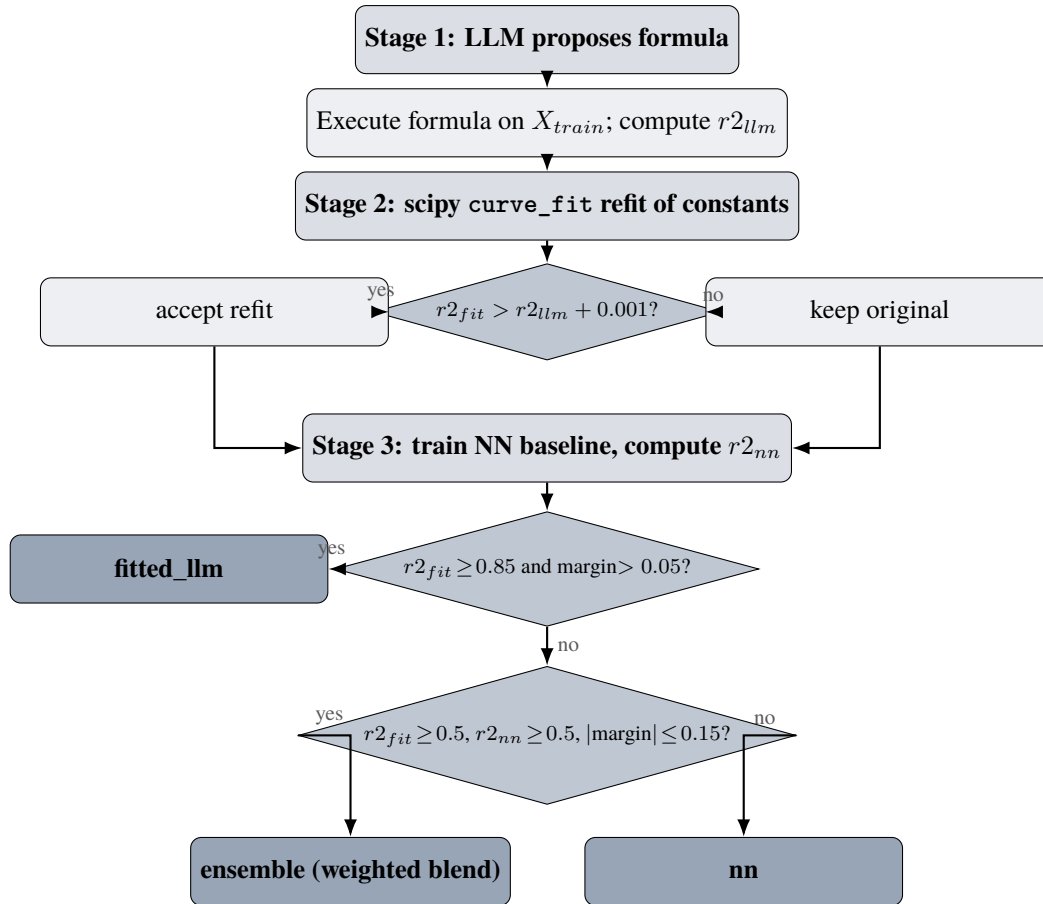
Note: `run_hybrid_system_benchmark.py`, `run_instability_suite.py` and the `portfolio_variance_v3c2.py/_v4c2.py` audit scripts were also read but showed no direct class-level reference to a specific variant at the top-level scope searched (they either re-import from the DeFi v3/v4 modules above or operate on already-produced result files), so they are omitted from Table 2 to avoid overstating the evidence.

Additional evidence source: the master orchestration script `run_all.sh` (uploaded separately, not in the master tree) lists the full pipeline as an ordered sequence of named steps (`exp1`, `exp1b`, `exp1_ablation`, `exp1_five`, `hybrid_all_domains`, `exp2_feynman`, `exp3`, `exp3b`, `suppA`, `suppB`, ...), each invoking one or more of the experiment scripts above with explicit paths and environment variables (e.g. `PYSR_POPULATIONS`, `METHOD_TIMEOUT`, `ANTHROPIC_API_KEY` implied by `exp1_ablation`’s own warning text). Two points from it materially strengthen Table 2:

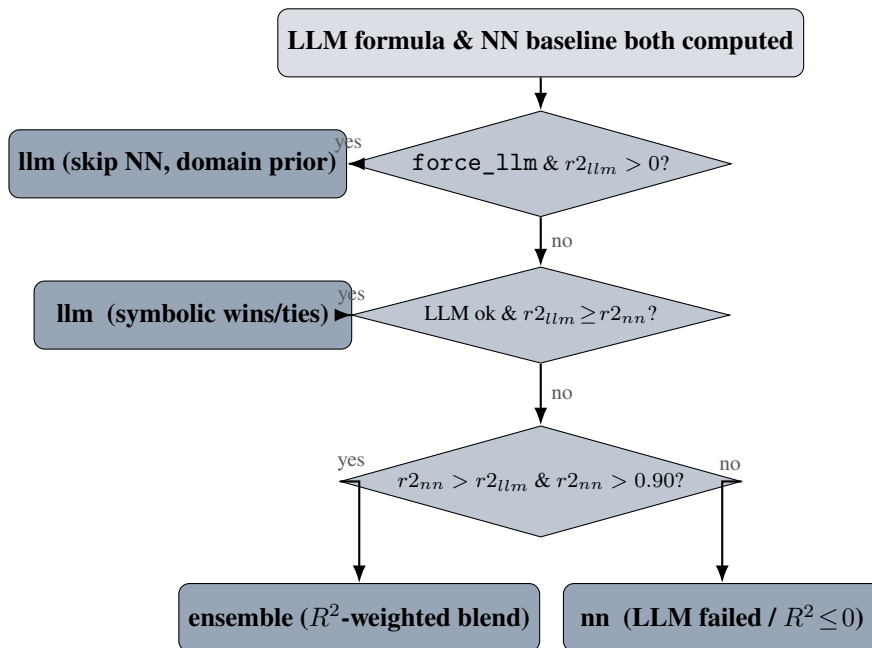
- the `hybrid_all_domains` step (§10.9 of the paper) cds directly into `hybrid_all_domains_llm_nn/` and runs `hybrid_system_llm_nn_all_domains.py` as a standalone script – i.e. Variant 2’s own file *is* the experiment, not merely imported by one;
- the `exp1/exp1b` steps run `hypatix_defi_benchmark_v4.py` directly as the paper’s core DeFi benchmark (Tab. 9–15), confirming Variant 1 (and, via its internal `_ensemble_llm_nn`, Variant 3) as production-path code rather than illustrative-only.

3 3) Algorithm diagrams

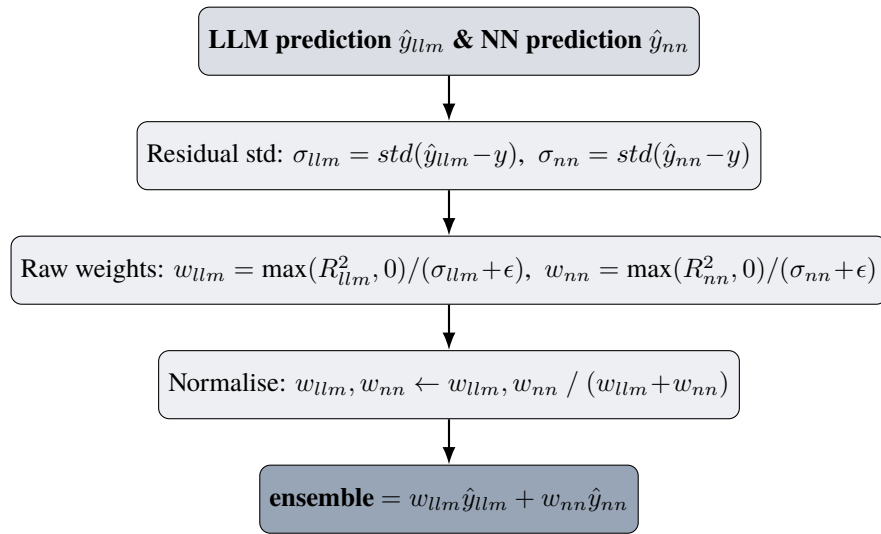
3.1 Variant 1 – Enhanced Hybrid DeFi (EnhancedHybridSystemDeFi)



3.2 Variant 2 – Hybrid All-Domains (HybridSystemAllDomains)

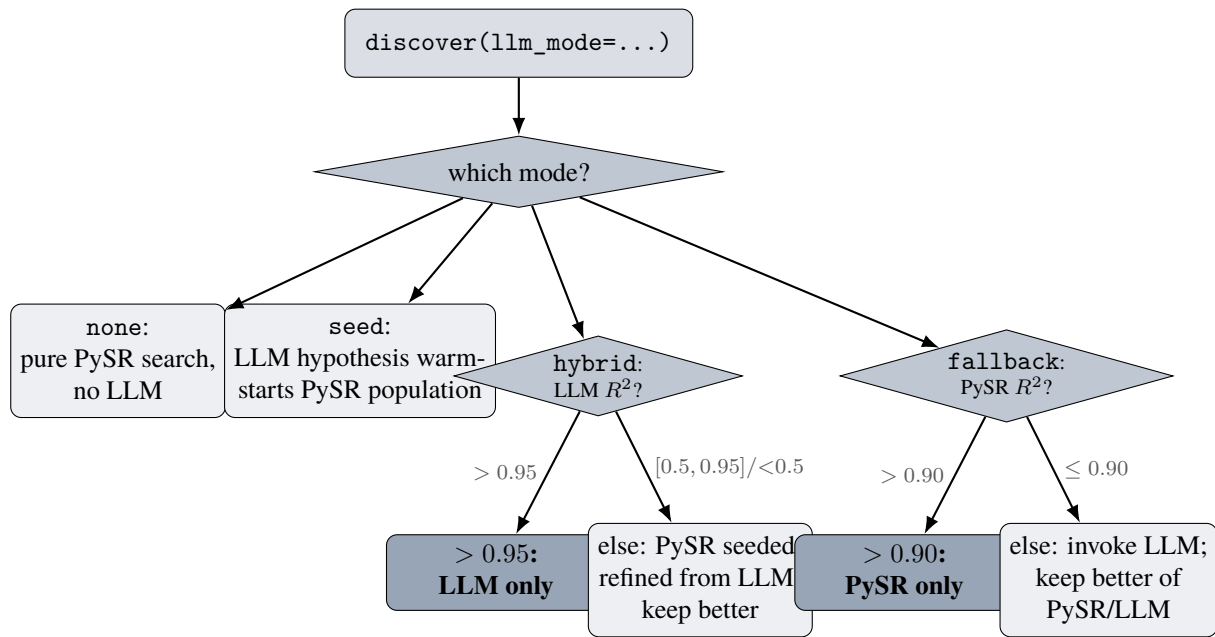


3.3 Variant 3 – Ensemble Example (ensemble_llm_nn)



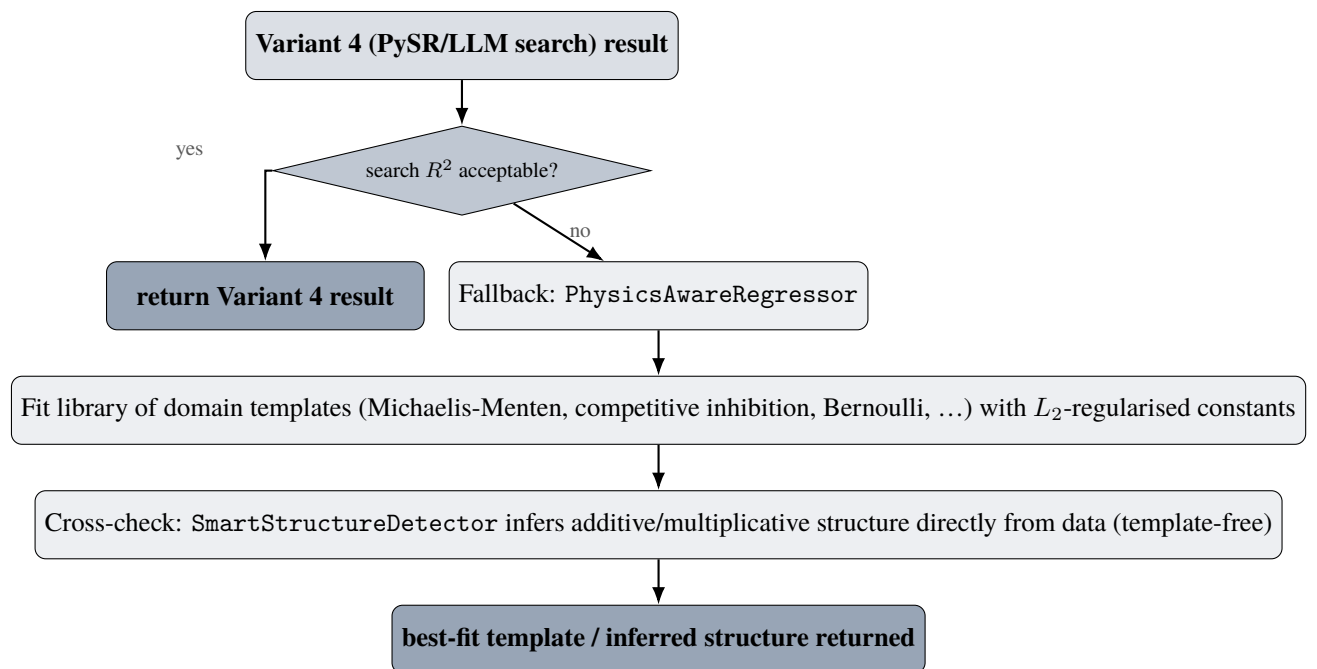
No hard decision gate: this variant always returns a precision-weighted blend of both models' predictions.

3.4 Variant 4 – LLM-Prior / PySR Family (SymbolicEngineWithLLM, mode-dispatched)



Orchestrated by HybridDiscoverySystem [T1], which also adds the Variant 5 fallback tier below when this whole family underperforms.

3.5 Variant 5 – Template Physics Fallback (PhysicsAwareRegressor / SmartStructureDetector)



4 Summary

Variant	Combination rule	Domain	Experiments
1. Enhanced Hybrid DeFi	threshold gate + scipy refit	DeFi	run_comparative_suite_*, run_noise_sweep_*, run_sample_complexity_*, test_enhanced_defi_*
2. Hybrid All-Domains	tie-favours-symbolic gate + force_llm override	Physics/Chem/Bio	run_comparative_suite_*, run_sample_complexity_*
3. Ensemble Example	inverse-residual-std weighting	DeFi (demo, but used)	hypatiaux_defi_benchmark_v3/v4[test_enhanced_defi_*
4. LLM-Prior / PySR family	4 modes: none/seed/hybrid/fallback	Nguyen/Feynman	hypatia.py, exp1_ablation, exp1_five_system, exp3_nguyen12_*, run_comparative_suite_*
5. Template physics fallback	template-library or template-free structure fit	Biology/Chem/Eng.	transitively via Variant 4's driver (exp1_ablation, run_comparative_suite_*)

This report was produced from static reading of the source files in <https://github.com/sednabcn/LLM-HypatiaX-REPRO> (fetched from master); no code in the repository was executed. Diagrams are faithful redrawings of the control flow as read from source, not renderings of any runtime trace.